

**AMRITA VISHWA VIDYAPEETHAM,
AMRITAPURI**

**B. Tech CSE (ARTIFICIAL
INTELLIGENCE)**

LITERATURE REVIEW SUBMISSION

Group No: 5

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1. Problem Definition

The research problem focuses on leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to extract, analyze, and represent structured knowledge from Ayurvedic texts written in Sanskrit. Classical texts like the Charaka Saṃhitā and Suśruta Saṃhitā are rich sources of medical knowledge but pose challenges due to complex linguistic constructs such as sandhi, samāsa, free word order, and the morphological richness of Sanskrit. The goal is to design an AI-based framework that performs both syntactic and semantic analysis of these texts, leading to the construction of a queryable Knowledge Graph (KG). This KG will include entities such as DRAVYA (substances/herbs), ROGA (diseases), LAKSHANA (symptoms), and DOSHA (bio-energetic principles), along with their inter-relationships. The ultimate objective is to bridge traditional Ayurvedic wisdom with modern computational tools for knowledge retrieval, decision support, and further research in integrative medicine.

2. Literature Review Summary Table

No.	Author(s) & Proceedings	Title	Year	Dataset Used	Methodology / Approach	Limitations & Future Work
1	Behr et al., RSC Catalysis Sci. Technol.	Knowledge Graphs in Chemistry: Ontology Learning and NER Pipelines	2024	Chemistry corpora	Review: NER + ontology learning + KG construction pipelines	Domain-agnostic; transferable methods only
2	Bagchi et al., RSUSCI Conf.	NLP-Based Knowledge Extraction from Charak Samhita for Navigating Ancient Wisdom	2024	OCR'd Charaka Saṃhitā	Django system: OCR + NER + query interface	Used translations, Sanskrit-native pipeline needed

3	Terdalkar & Bhattacharya, arXiv	Automated KG Creation and QA from Sanskrit Epics	2023	Mahābhārata, Rāmāyaṇa, Bhāvaparakāśa synsets	Automated KG creation + QA in Sanskrit (includes Ayurveda terms)	Early-stage, needs full Ayurvedic ontology
4	Kulkarni et al., Sanskrit NLP Workshop	Advances in Morphological Segmentation and Compound Analysis in Sanskrit	2022	Sanskrit Digital Corpus	Morpho-syntactic parsing for Sanskrit	No Ayurvedic integration; useful for preprocessing
5	Kulkarni et al., WSC Proceedings	Sanskrit Semantic Parsing for Knowledge Extraction	2023	Sanskrit annotated datasets	Semantic parsing for extracting relations	Not domain-specific; Ayurveda corpus missing

3. Review of Selected Papers

Behr et al. (2024) presented a comprehensive review of knowledge graphs in chemistry, covering ontology learning, named entity recognition (NER), and KG construction pipelines. While their methods are domain-agnostic, they highlight reusable techniques that can be adapted to Ayurveda.

Bagchi et al. (2024) developed an NLP-based knowledge extraction system for the Charaka Saṃhitā. Using OCR, NER, and a Django-based query interface, they demonstrated end-to-end information retrieval. However, their reliance on translations underscores the need for a Sanskrit-native pipeline.

Terdalkar and Bhattacharya (2023) explored automated KG creation and question answering for Sanskrit epics such as the Mahābhārata and Rāmāyaṇa, with extensions into Ayurvedic texts like the Bhāvaparakāśa. Their work demonstrates the feasibility of automated Sanskrit KG construction but remains at an early stage, requiring a comprehensive Ayurvedic ontology for scalability.

Kulkarni et al. (2022) advanced morphological segmentation and compound analysis in Sanskrit. Their morpho-syntactic parsing methods are particularly valuable for handling sandhi and samāsa structures, making them highly relevant as preprocessing tools for Ayurvedic corpora.

Kulkarni et al. (2023) focused on semantic parsing for knowledge extraction from Sanskrit annotated datasets. Their work effectively extracts relations between entities but lacks application to Ayurveda, emphasizing the gap in domain-specific corpora.

4. Identified Research Gaps

- No comprehensive Ayurvedic ontology covering key concepts like dravya, roga, lakṣaṇa, and doṣa.
- Most systems rely on translations, reducing accuracy and domain-specific precision.
- Manual effort is still heavily involved in KG creation, requiring automation.
- Existing NLP tools for Sanskrit are domain-agnostic, lacking Ayurveda-specific adaptation.
- There are no Ayurvedic-specific QA systems for answering questions based on Ayurvedic knowledge.

5. Novelty / Contribution of Proposed Work

- Automates creation of Knowledge Graphs from Sanskrit texts.
- Sanskrit-Specific NLP: Uses tailored grammar rules and heuristics for Sanskrit.
- Cross-Domain Application: Applies to epic relationships and Ayurvedic synonyms.
- Question-Answering System Integrates a tool to query the Knowledge Graph.
- Develop a domain-specific QA system for Ayurvedic KGs.

Date: 30 August 2025

Sign: _____